Correlating Muon Flux and Lightning Occurrence During a Thunderstorm

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~ Introduction ~

High energy particle correlations with lightning initiation is poorly understood [4]. Project BoTTLE (Bounded Thunderstorm Tracking of Lightning Events with muon detection) utilizes muon detectors to systematically monitor muon particle activity in conjunction with observed lightning events. Our objective is to examine whether significant correlations exist between decreasing muon flux and the occurrence of lightning discharges.

Key Words: Muon-Flux, Lightning Counts, Atmospheric Dynamics

~ Background ~

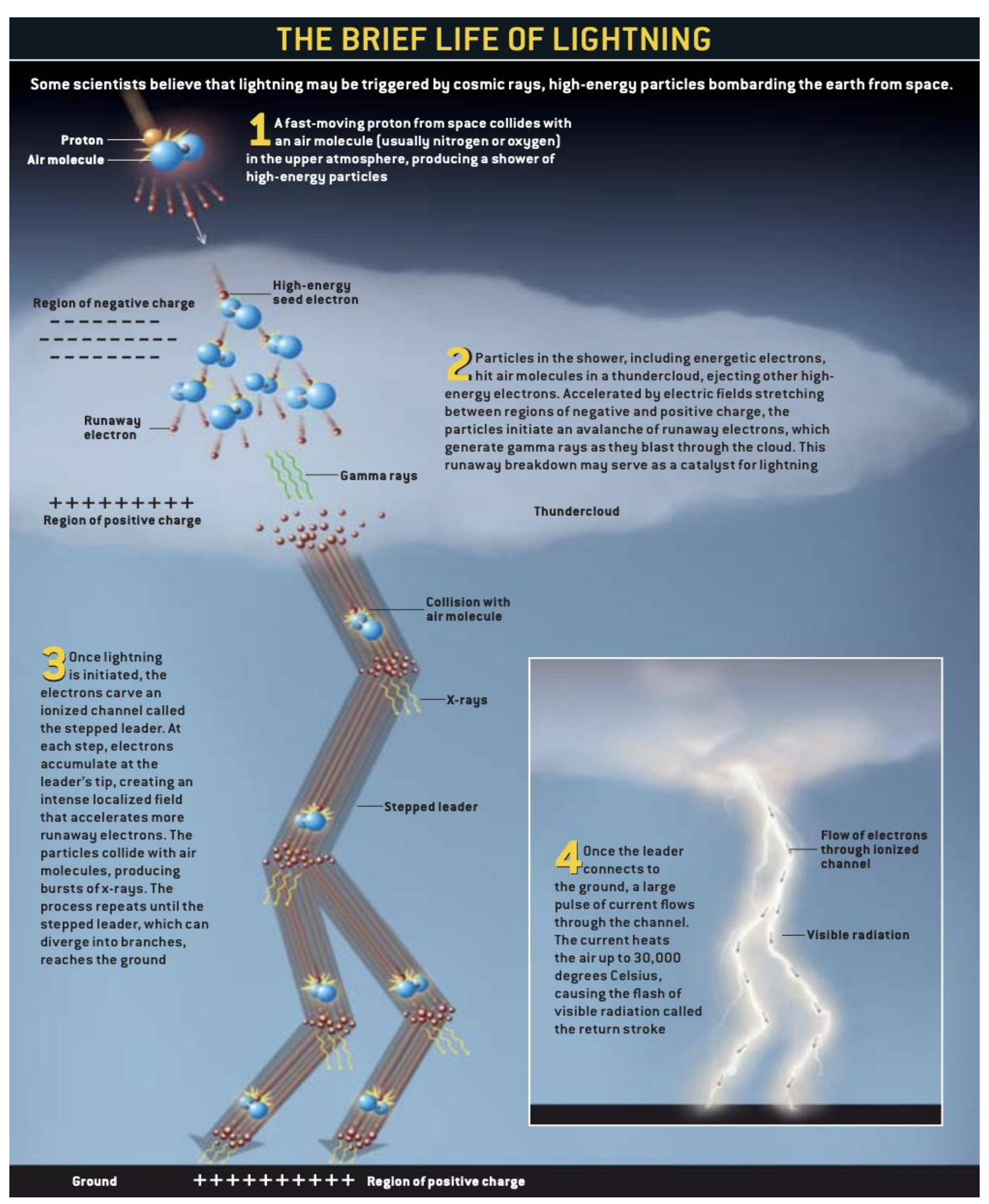
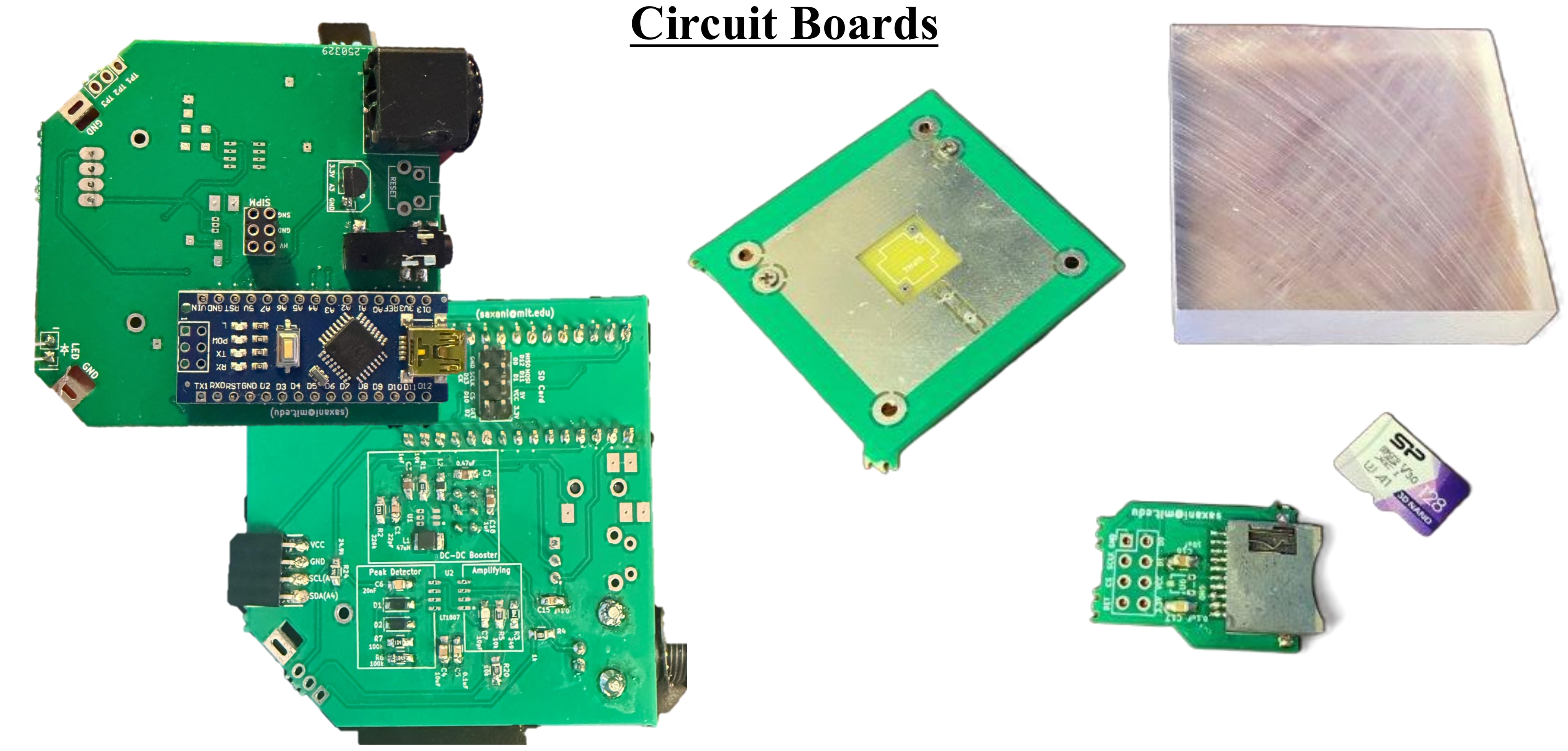


Figure 1: From the 2005 Scientific American Magazine, Joseph R Dwyer provides a great comprehensive image of lightning phenomena, highlighting our motivation for muon detections.

- The muon is a high-energy particle from cosmic rays and indicates upper-atmospheric particle cascades and energetic processes [2].
- Muons serve as a probe for investigating lightning, since changes in muon flux can measure potential within a storm [2].
- Lightning is a physical phenomenon whose governing dynamics remain unknown despite extensive research [4].
- We chose muons as our primary detection source due to their abundance in Earth's upper atmosphere and the high-energy nature of cosmic rays [1, 3].

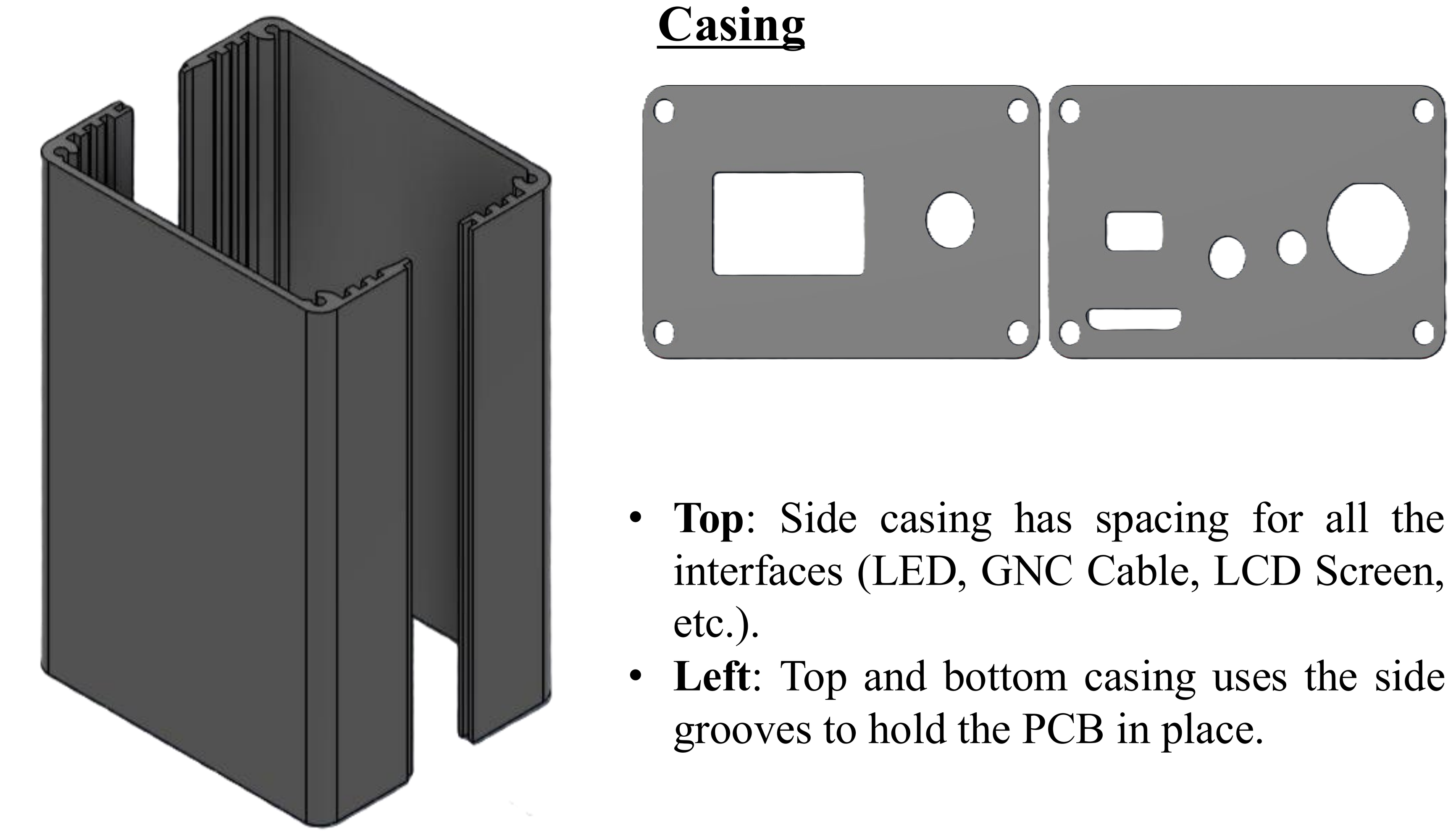
~ Muon Detectors ~

Circuit Boards



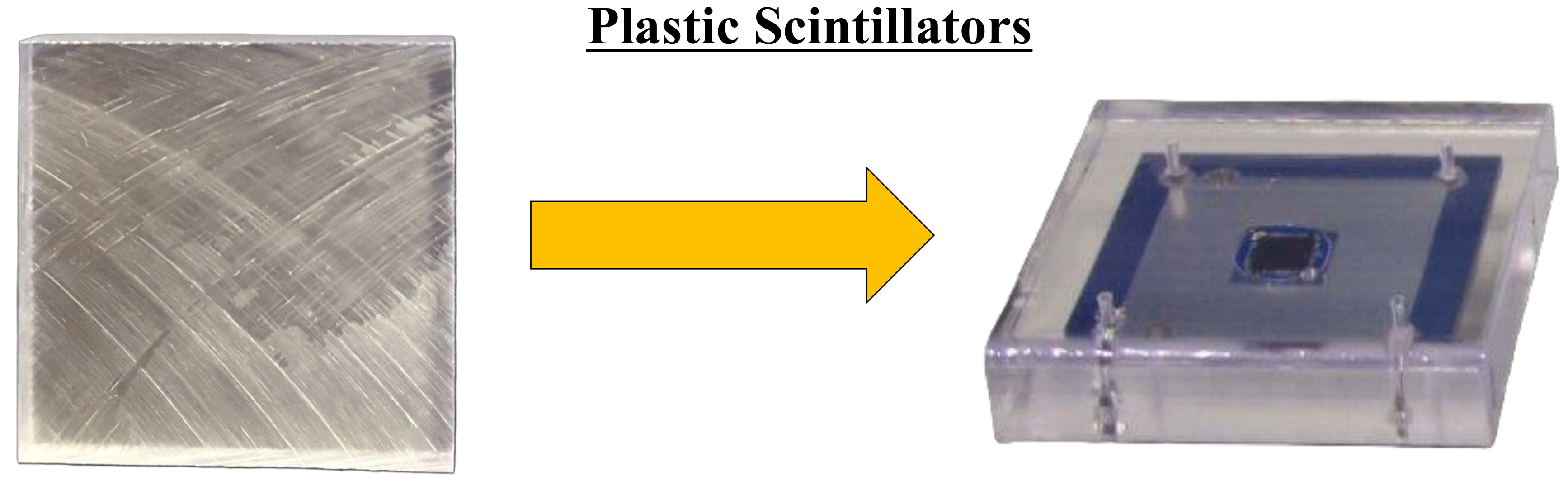
- Plastic scintillators emit light at specific wavelengths.
- Photons entering the detector pass through an **SiPM**, where an ionization cascade produces a signal that is then read by the Arduino nano.
- **Left**: Example PCB board(s) for data processing and interfacing front and backside
- **Right**: SD card for memory storage, SiPM, and scintillator

Casing



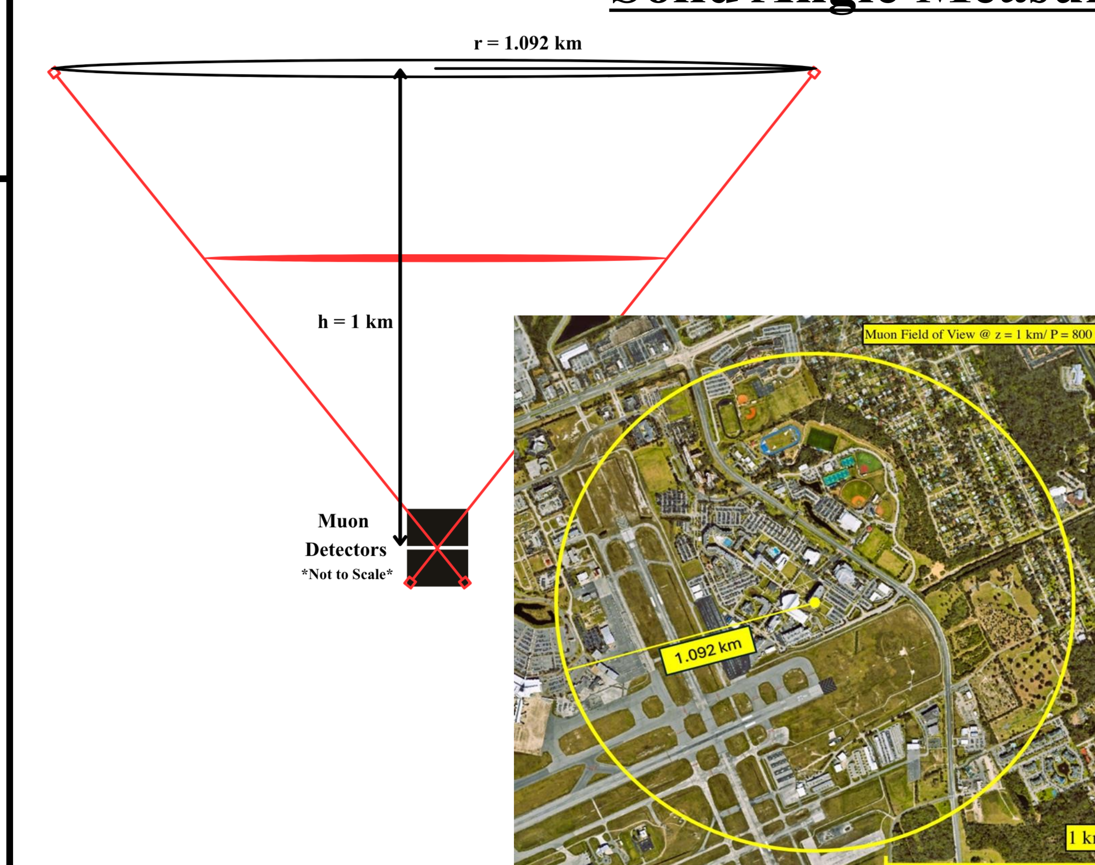
- **Top**: Side casing has spacing for all the interfaces (LED, GNC Cable, LCD Screen, etc.).
- **Left**: Top and bottom casing uses the side grooves to hold the PCB in place.

Plastic Scintillators



- **Left**: Scintillator prior to sanding, drilling, and polishing.
- **Right**: Scintillator post polishing and attached to the SiPM.

Solid Angle Measurement



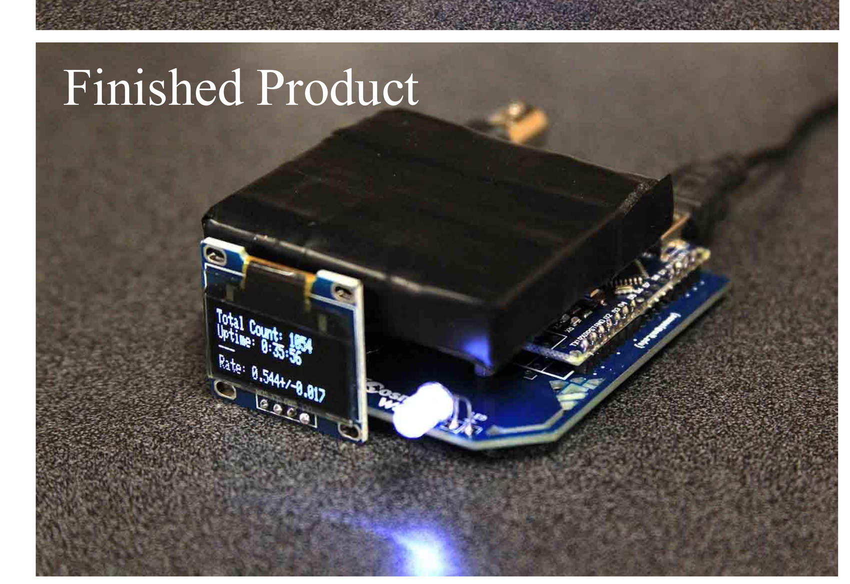
- The muon detectors can be stacked to enable more precise measurements and the derivation of a solid angle for the spatial coverage. The main variable is the distance between the scintillators [2].
- Calculated spatial coverage for the muon detectors is approximately 3.75 km^2 at a height of 1 km with a muon incidence angle of $\pm$ 47.53º.

~ Results & Remarks ~

Next Steps:

- Evaluate statistical validity of muon detector measurements, increased detector coverage, and integration with atmospheric measurements to better constrain the physical relationship between muon influx and lightning events.
- We will conduct testing on the roof of the College of Arts and Sciences (COAS). Current testing trials are included below:

Distance Between Scintillators [cm]	Muon Incidence Angle [º]	Spatial Coverage at 1 km Altitude [km^2]
3.5 (minimum)	± 47.53	3.75
7.0	± 23.25	0.58
14	± 11.56	0.13
28	± 5.77	0.03
56	± 2.89	0.008
112	± 1.44	0.002



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